WORKED SOLUTION

STEP I 2017, Question 10

Mechanics • Colliding particles

STEP I 2017, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q10

The key idea

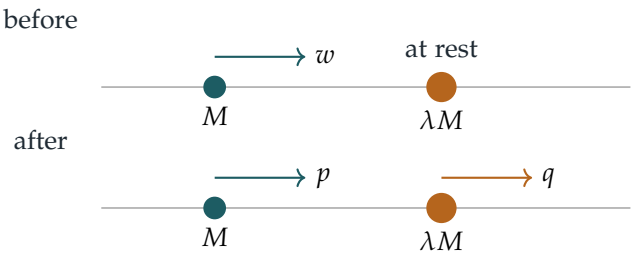
Every collision in the chain looks the same: a particle of mass M moving at speed w hits a particle of mass λM at rest. So we solve that one collision once, in terms of the ratio λ, and the whole chain follows by repeating it, giving geometric sequences. After that, each part is a matter of comparing the velocities the particles are left with, to see whether anyone catches up with the particle in front.

Solution

Model all the particles as points moving along the x-axis, with velocities measured in the positive x-direction.

One typical collision

A particle A of mass M moving at speed w hits a particle B of mass λM at rest. Let their velocities afterwards be p and q.



Conservation of momentum and Newton's experimental law give

$$Mw = Mp + \lambda Mq, \qquad q - p = ew.$$

Substituting $p = q - ew$ into the first equation, $w = q - ew + \lambda q$, so

$$q = \frac{1+e}{1+\lambda}\,w, \qquad p = q - ew = \frac{1-\lambda e}{1+\lambda}\,w.$$

Write $r = \dfrac{1+e}{1+\lambda}$ and $s = \dfrac{1-\lambda e}{1+\lambda}$. Note that

$$q - p = ew > 0,$$

so straight after any such collision the two particles separate.

Part (i)

The first collision is P (mass m, speed u) hitting P_1 (mass λm). Taking $w = u$ and $M = m$,

$$u_1 = \frac{1+e}{1+\lambda}\,u,$$

as required, and P is left with velocity su.

P_1 then moves off and hits P_2, of mass $\lambda^2 m = \lambda \cdot (\lambda m)$. The mass ratio is again λ, so this is the same collision with $w = u_1$. In general P_n (mass $\lambda^n m$) hits P_{n+1} (mass $\lambda^{n+1} m$) at speed u_n, so $u_{n+1} = ru_n$ and P_n is left with velocity su_n. Hence

$$u_n = \left(\frac{1+e}{1+\lambda}\right)^n u.$$

Here the first collision of P_n is being hit by P_{n-1}, and its second is hitting P_{n+1}. (Straight after its first collision P_n is moving away from P_{n-1}, so the next thing that can happen to it is reaching P_{n+1}, provided nothing behind has caught it up first. This is the chain of events the question intends, and in parts (ii)–(iv) we check it really happens.) So

$$v_n = su_n = \frac{(1-\lambda e)(1+e)^n}{(1+\lambda)^{n+1}}\,u,$$

as a velocity in the positive x-direction. As a speed it is the modulus of this, which matters only if $\lambda e > 1$, when P_n bounces back.

Part (ii)

Let $e > \lambda$. Then $\lambda < e < 1$, so $\lambda e < 1$ and $s > 0$; also $r > 1$.

The collisions take place in order: P with P_1, then P_1 with P_2, then P_2 with P_3, and so on. After P_n has struck P_{n+1}, the velocities from the back of the line are

$$su,\ su_1,\ su_2,\ \ldots,\ su_n,\ u_{n+1},\ 0,\ 0,\ \ldots$$

for $P, P_1, \ldots, P_n, P_{n+1}, P_{n+2}, \ldots$. Since $u_k = r^k u$ with $r > 1$ and $s > 0$,

$$0 < su < su_1 < su_2 < \cdots < su_n,$$

and $u_{n+1} > su_n$ because $u_{n+1} - su_n = (r-s)u_n = eu_n > 0$. So among $P, P_1, \ldots, P_{n+1}$, each particle is moving strictly faster than the one behind it, and no two of them can

ever meet again. The only collision still to come is P_{n+1} reaching the stationary P_{n+2}, and the same argument then applies with $n+1$ in place of n.

So each P_n ($n \geq 1$) is hit once by P_{n-1}, then hits P_{n+1} once, and is never involved in another collision: exactly two collisions each. (This also confirms the order of events assumed in (i).)

Part (iii)

If $e = \lambda$ then $r = 1$ and $s = \dfrac{1-e^2}{1+e} = 1-e$. So

$$u_n = u \quad \text{and} \quad v_n = (1-e)u \text{ for every } n,$$

and P is also left with velocity $(1-e)u$. The motion is a pulse travelling along the line: the particle at the front always moves at speed u and, on hitting the next particle, hands on speed u and drops to $(1-e)u$. Everything behind moves together at $(1-e)u$, so no further collisions happen among them, and the particle in front, at speed u, pulls away from them. Each P_n is involved in exactly two collisions.

After N collisions, $P, P_1, \ldots, P_{N-1}$ move at $(1-e)u$ and P_N (mass $\lambda^N m = e^N m$) at u. The kinetic energy is

$$\begin{aligned}\tfrac{1}{2}m(1-e)^2u^2(1+e+\cdots+e^{N-1}) + \tfrac{1}{2}e^N mu^2 &= \tfrac{1}{2}mu^2\left[(1-e)^2\,\frac{1-e^N}{1-e} + e^N\right] \\ &= \tfrac{1}{2}mu^2[(1-e)(1-e^N)+e^N] \\ &= \tfrac{1}{2}mu^2[1-e+e^{N+1}].\end{aligned}$$

Dividing by the initial energy $\frac{1}{2}mu^2$, the fraction lost after N collisions is

$$1-(1-e+e^{N+1}) = e(1-e^N) \to e \quad \text{as } N \to \infty,$$

since $0 < e < 1$. This is the required result.

Part (iv)

If $\lambda e = 1$ then $s = 0$ and

$$r = \frac{1+e}{1+1/e} = \frac{e(1+e)}{e+1} = e.$$

So every collision brings the moving particle dead to rest: P stops, then P_1 moves off at eu, hits P_2 and stops, and so on, with $u_n = e^n u$ and $v_n = 0$. Each P_n has exactly two collisions, and the particles end up at rest, each touching the next, while a single particle carries the motion forward ever more slowly. As P_n takes time $1/(e^n u)$ to reach P_{n+1}, the gaps between collisions grow without bound.

Ask yourself Momentum is conserved, so how can the energy go?

After N collisions only P_N moves, with mass $\lambda^N m = e^{-N}m$ and speed $e^N u$. Its momentum is $e^{-N}m \cdot e^N u = mu$, as it must be, but its kinetic energy is

$$\tfrac{1}{2}\,e^{-N}m\,(e^N u)^2 = e^N \cdot \tfrac{1}{2}mu^2.$$

The momentum is carried by ever heavier, ever slower particles, and kinetic energy $p^2/(2M)$ for fixed momentum p tends to zero as the mass M grows. The fraction remaining, e^N, tends to 0, so eventually *all* of the initial kinetic energy is lost.

Checks

- With $e = 1$ and $\lambda = 1$ (equal masses, perfectly elastic), $u_1 = u$ and P stops: the familiar Newton's cradle exchange.
- In (iii), a single collision ($N = 1$) loses the fraction $e(1 - e)$. Directly: P ends at $(1 - e)u$ and P_1, of mass em, at u, so the energy left is $(1 - e)^2 + e = 1 - e + e^2$ of the original, which agrees.
- A step-by-step computer simulation of the collisions, with exact fractions, confirmed the formulae for u_n and v_n, the two-collisions-each pattern for $e > \lambda$, the fraction lost $e(1 - e^N)$ (for $e = \frac{1}{2}$, 0.49999998 after 25 collisions) and the remaining fraction e^N when $\lambda e = 1$.

Common slips

- Writing Newton's law as $q - p = -ew$ or the wrong way round: the speed of separation is e times the speed of approach, and here both are positive.
- Forgetting that the mass ratio is λ at *every* collision, which is what makes the answers geometric.
- Assuming the order of events without checking it. When $e < \lambda$ and $\lambda e < 1$, a particle left behind can catch up with the one in front before that one reaches the next particle, and the simple pattern breaks down.

Beyond STEP

Part (iv) shows momentum and energy behaving very differently: momentum is conserved exactly, yet the energy drains away completely because it is carried by ever heavier particles. Chains of colliding particles are a simple model of how a pulse travels through a medium of varying density, and they lead on to wave propagation in discrete lattices. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions and the centre-of-mass frame with this kind of care.

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